Process Design Project – Reactor and Separation System

Overview

This project involved the analysis and modelling of a steady-state chemical process consisting of a reactor followed by a separation unit. The objective was to determine the flow rates of components through the system and evaluate the effect of conversion on product formation.

Process Description

A feed stream of 100 kmol/h of component A enters a reactor, where it undergoes a chemical reaction converting A to B. The reactor operates at a conversion of 60%. The outlet stream is then sent to a separator, where 90% of component B is recovered as the desired product, while the remainder is removed as waste

Methodology

Steady-state material balance equations were applied to each unit in the system. For the reactor, conversion-based relationships were used to determine outlet flow rates of reactants and products. The separator was modelled using a fixed recovery fraction for component B.

An Excel model was developed to perform the calculations and allow easy modification of input parameters such as conversion and separation efficiency. Python was used to visualize the relationship between conversion and product formation.

Results

- Outlet flow of A: 40 kmol/h
- Outlet flow of B: 60 kmol/h
- Product stream flow rate: 94 kmol/h
- Product B recovered: 54 kmol/h
- Waste stream flow rate: 6 kmol/h

The results demonstrate the direct relationship between conversion and product generation, highlighting the importance of reactor performance in process efficiency.

Tools Used

- Microsoft Excel (process modelling and calculations)
- Python (data visualization using Matplotlib)

Conclusion

This project demonstrates the application of fundamental material balance principles to a simple process system. It highlights how conversion and separation efficiency influence overall production and provides a basic framework for process design and analysis.